THE KUMMER PAIRING AND WEAK MORDELL–WEIL

SANTIAGO ARANGO-PIÑEROS

The weak Mordell–Weil theorem says that $E(K)/mE(K)$ is finite for an elliptic curve E over a number field K. The proof in [Sil09, §VIII.1] has three steps. The Kummer pairing turns the finiteness of $E(K)/mE(K)$ into the finiteness of the field L generated by the m-th divisors of the rational points. This field is abelian of exponent m and unramified outside a finite set of places S. And Kummer theory shows that such fields are finite. This worksheet runs the argument for $m = 2$ and $K = \mathbb{Q}$, on curves whose points of order two are rational. Then the class group and the S-unit theorem disappear, and the injectivity of reduction on the 2-torsion becomes elementary. Problem 1.6 sets up the pairing. Problem 2.7, the heart of the worksheet, proves that L is unramified outside S and deduces a bound for $E(\mathbb{Q})/2E(\mathbb{Q})$. Problem 3.1 tests the bound on $y^2 = x^3 - x$. Each problem is preceded by the definitions and results it uses.

Standing notation.

- An elliptic curve E over $\mathbb{Q}$, given by
$$y^2 = (x - e_1)(x - e_2)(x - e_3)$$
with distinct $e_1, e_2, e_3 \in \mathbb{Z}$, and its points of order two $T_i := (e_i, 0)$. So $E[2](\overline{\mathbb{Q}}) = \{O, T_1, T_2, T_3\} \subseteq E(\mathbb{Q})$.
- The finite set of primes $S := \{2\} \cup \{p : p \mid (e_1 - e_2)(e_1 - e_3)(e_2 - e_3)\}$, and $s := \#S$.
- The absolute Galois group $\mathrm{Gal}_{\mathbb{Q}} := \mathrm{Gal}(\overline{\mathbb{Q}}/\mathbb{Q})$, with elements σ, τ, γ. For an extension F of $\mathbb{Q}$ inside $\overline{\mathbb{Q}}$, $\mathrm{Gal}_F := \mathrm{Gal}(\overline{\mathbb{Q}}/F)$.
- For $Q \in E(\overline{\mathbb{Q}})$, the field $\mathbb{Q}(Q)$ generated by the coordinates of Q, and the point $\sigma(Q)$, obtained by applying $\sigma \in \mathrm{Gal}_{\mathbb{Q}}$ to the coordinates.
- The compositum L of the fields $\mathbb{Q}(Q)$ for $Q \in E(\overline{\mathbb{Q}})$ with $2Q \in E(\mathbb{Q})$. Silverman writes $L = \mathbb{Q}([2]^{-1}E(\mathbb{Q}))$.
- The quotient $B/2 := B/2B$ of an abelian group B, and the group $\mathbb{Q}^\times/\mathbb{Q}^{\times 2}$ of classes of rational numbers modulo squares.

How to use this. Allow about 20 minutes for Problem 1.6, 35 for Problem 2.7 and 25 for Problem 3.1. Problem 2.7 is the one not to skip. If cohomology has not come up yet in lecture, leave Problem 1.6(4) for later. The `Magma` block runs in seconds in the free online calculator. Everything else is by hand.

1. The Kummer pairing

This section is [Sil09, Proposition VIII.1.2], for $m = 2$ and $K = \mathbb{Q}$, and its reinterpretation in [Sil09, §VIII.2], with cohomology as in [Sil09, §§B.1–B.2].

Definition 1.1 (The Kummer pairing [Sil09, §VIII.1])**.** The Kummer pairing is the map
$$\kappa\colon E(\mathbb{Q}) \times \mathrm{Gal}_{\mathbb{Q}} \to E[2](\overline{\mathbb{Q}})$$
defined as follows. Let $P \in E(\mathbb{Q})$ and $\sigma \in \mathrm{Gal}_{\mathbb{Q}}$. Choose $Q \in E(\overline{\mathbb{Q}})$ with $2Q = P$, and set $\kappa(P, \sigma) := \sigma(Q) - Q$. That this lands in $E[2](\overline{\mathbb{Q}})$, and does not depend on Q, is Problem 1.6(1).

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Proposition 1.2 (Infinite Galois theory [Mil22, Theorem 7.13])**.** *For a closed subgroup H of $\mathrm{Gal}_{\mathbb{Q}}$, let $\overline{\mathbb{Q}}^H$ be its fixed field. The maps $H \mapsto \overline{\mathbb{Q}}^H$ and $F \mapsto \mathrm{Gal}_F$ are inverse bijections between the closed subgroups of $\mathrm{Gal}_{\mathbb{Q}}$ and the extensions of $\mathbb{Q}$ inside $\overline{\mathbb{Q}}$. The subgroup H is normal if and only if $\overline{\mathbb{Q}}^H$ is Galois over $\mathbb{Q}$, and then $\mathrm{Gal}(\overline{\mathbb{Q}}^H/\mathbb{Q}) \cong \mathrm{Gal}_{\mathbb{Q}}/H$ by restriction. In particular, $\overline{\mathbb{Q}}^{\mathrm{Gal}_{\mathbb{Q}}} = \mathbb{Q}$.*

Definition 1.3 (Cohomology in degree 1 [Sil09, §§B.1–B.2])**.** Let A be a discrete $\mathrm{Gal}_{\mathbb{Q}}$-module. A 1-cocycle is a continuous map $\xi\colon \mathrm{Gal}_{\mathbb{Q}} \to A$, $\sigma \mapsto \xi_\sigma$, with $\xi_{\sigma\tau} = \xi_\sigma + \sigma\xi_\tau$ for all σ, τ. For $a \in A$, the map $\sigma \mapsto \sigma a - a$ is a 1-cocycle, and the 1-cocycles of this form are the 1-coboundaries. Then $H^1(\mathrm{Gal}_{\mathbb{Q}}, A)$ is the group of 1-cocycles modulo the 1-coboundaries. For an elliptic curve or its 2-torsion, we abuse notation and write $H^1(\mathbb{Q}, E[2]) \coloneqq H^1(\mathrm{Gal}_{\mathbb{Q}}, E[2](\overline{\mathbb{Q}}))$.

Proposition 1.4 (The long exact sequence [Sil09, Propositions B.1.2 and B.2.3])**.** *Let $0 \to A \to B \xrightarrow{\psi} C \to 0$ be an exact sequence of discrete $\mathrm{Gal}_{\mathbb{Q}}$-modules. For $c \in C^{\mathrm{Gal}_{\mathbb{Q}}}$, choose $b \in B$ with $\psi(b) = c$, and let $\delta(c) \in H^1(\mathrm{Gal}_{\mathbb{Q}}, A)$ be the class of $\sigma \mapsto \sigma b - b$. This does not depend on b, and the sequence*

$$0 \to A^{\mathrm{Gal}_{\mathbb{Q}}} \to B^{\mathrm{Gal}_{\mathbb{Q}}} \to C^{\mathrm{Gal}_{\mathbb{Q}}} \xrightarrow{\delta} H^1(\mathrm{Gal}_{\mathbb{Q}}, A) \to H^1(\mathrm{Gal}_{\mathbb{Q}}, B)$$

is exact.

Proposition 1.5 (Trivial action [Sil09, Remark B.2.2])**.** *If $\mathrm{Gal}_{\mathbb{Q}}$ acts trivially on A, then*

$$H^1(\mathrm{Gal}_{\mathbb{Q}}, A) = \mathrm{Hom}_{\mathrm{conts}}(\mathrm{Gal}_{\mathbb{Q}}, A),$$

the group of continuous homomorphisms.

Problem 1.6 (The Kummer pairing)**.**

(1) Show that $\kappa(P, \sigma) \in E[2](\overline{\mathbb{Q}})$, and that it does not depend on the choice of Q.
(2) Let $P, P' \in E(\mathbb{Q})$ and $\sigma, \tau \in \mathrm{Gal}_{\mathbb{Q}}$. Show that $\kappa(P + P', \sigma) = \kappa(P, \sigma) + \kappa(P', \sigma)$ and

$$\kappa(P, \sigma\tau) = \kappa(P, \sigma) + \sigma\big(\kappa(P, \tau)\big).$$

Deduce that κ is bilinear. Where does the hypothesis $E[2](\overline{\mathbb{Q}}) \subseteq E(\mathbb{Q})$ enter?
(3) Show that $\kappa(P, \cdot) = O$ if and only if $P \in 2E(\mathbb{Q})$, and that $\kappa(\cdot, \sigma) = O$ if and only if $\sigma \in \mathrm{Gal}_L$. Deduce that $L/\mathbb{Q}$ is Galois and that κ induces a perfect pairing

$$E(\mathbb{Q})/2 \times \mathrm{Gal}(L/\mathbb{Q}) \to E[2](\overline{\mathbb{Q}}),$$

that is, both $E(\mathbb{Q})/2 \to \mathrm{Hom}(\mathrm{Gal}(L/\mathbb{Q}), E[2](\overline{\mathbb{Q}}))$ and $\mathrm{Gal}(L/\mathbb{Q}) \to \mathrm{Hom}(E(\mathbb{Q})/2, E[2](\overline{\mathbb{Q}}))$ are injective. Conclude that $E(\mathbb{Q})/2$ is finite if and only if $L/\mathbb{Q}$ is finite.
(4) Show that $0 \to E[2](\overline{\mathbb{Q}}) \to E(\overline{\mathbb{Q}}) \xrightarrow{2} E(\overline{\mathbb{Q}}) \to 0$ is an exact sequence of discrete $\mathrm{Gal}_{\mathbb{Q}}$-modules, and that its connecting map $\delta\colon E(\mathbb{Q}) \to H^1(\mathbb{Q}, E[2])$ sends P to the class of $\kappa(P, \cdot)$. Show that $H^1(\mathbb{Q}, E[2]) = \mathrm{Hom}_{\mathrm{conts}}(\mathrm{Gal}_{\mathbb{Q}}, E[2](\overline{\mathbb{Q}}))$, and recover from Proposition 1.4 the injectivity of $E(\mathbb{Q})/2 \to \mathrm{Hom}(\mathrm{Gal}_{\mathbb{Q}}, E[2](\overline{\mathbb{Q}}))$.

2. Unramified outside S

This section is [Sil09, Proposition VIII.1.5], for $m = 2$ and $K = \mathbb{Q}$. Its argument is that of [Sil09, Proposition VII.4.1(a)], applied to the halves of rational points instead of torsion points. The last item adds Kummer theory, as in [Sil09, proof of Proposition VIII.1.6]. In this section p is a prime, and $\overline{\mathbb{Q}}$ is embedded in an algebraic closure $\overline{\mathbb{Q}}_p$ of $\mathbb{Q}_p$, so that restriction is an injection $\mathrm{Gal}_{\mathbb{Q}_p} \coloneqq \mathrm{Gal}(\overline{\mathbb{Q}}_p/\mathbb{Q}_p) \hookrightarrow \mathrm{Gal}_{\mathbb{Q}}$. The residue field of the valuation ring of $\overline{\mathbb{Q}}_p$ is an algebraic closure $\overline{\mathbb{F}}_p$ of $\mathbb{F}_p$, and a bar denotes reduction modulo its maximal ideal.

Definition 2.1 (Inertia [Sil09, §VII.4])**.** An element $\sigma \in \mathrm{Gal}_{\mathbb{Q}_p}$ preserves the valuation ring of $\overline{\mathbb{Q}}_p$, so it induces an automorphism $\bar{\sigma}$ of $\overline{\mathbb{F}}_p$. The inertia group $I_p \subseteq \mathrm{Gal}_{\mathbb{Q}_p}$ is the subgroup of the σ with $\bar{\sigma} = 1$. If ${\mathbb{Q}_p}^{\mathrm{unr}} \subseteq \overline{\mathbb{Q}}_p$ is the maximal unramified extension of $\mathbb{Q}_p$, then $I_p = \mathrm{Gal}(\overline{\mathbb{Q}}_p/{\mathbb{Q}_p}^{\mathrm{unr}})$. A Galois extension F of $\mathbb{Q}$ inside $\overline{\mathbb{Q}}$ is unramified at p if $I_p \subseteq \mathrm{Gal}_F$, that is, if I_p fixes F.

Proposition 2.2 (Good reduction [Sil09, §III.1 and Remark VIII.1.3])**.** *The Weierstrass equation of E has discriminant*

$$\Delta = 16(e_1 - e_2)^2(e_1 - e_3)^2(e_2 - e_3)^2.$$

If $p \nmid \Delta$, the equation is minimal at p and its reduction E_p is an elliptic curve over $\mathbb{F}_p$.

Proposition 2.3 (Reduction [Sil09, Proposition VII.2.1])**.** *Suppose that $p \nmid \Delta$. Every $Q \in E(\overline{\mathbb{Q}}_p)$ has projective coordinates $[X : Y : Z]$ in the valuation ring of $\overline{\mathbb{Q}}_p$, not all in its maximal ideal, and $\bar{Q} := [\bar{X} : \bar{Y} : \bar{Z}] \in E_p(\overline{\mathbb{F}}_p)$. The reduction map $E(\overline{\mathbb{Q}}_p) \to E_p(\overline{\mathbb{F}}_p)$, $Q \mapsto \bar{Q}$, is a homomorphism, and $\overline{\sigma(Q)} = \bar{\sigma}(\bar{Q})$ for $\sigma \in \mathrm{Gal}_{\mathbb{Q}_p}$.*

Proposition 2.4 (Kummer theory [Sil09, Exercise 8.4])**.** *Let F be a Galois extension of $\mathbb{Q}$ inside $\overline{\mathbb{Q}}$ with $\mathrm{Gal}(F/\mathbb{Q})$ abelian of exponent dividing 2. Let $D \subseteq \mathbb{Q}^\times/\mathbb{Q}^{\times 2}$ be the subgroup of the classes of the $d \in \mathbb{Q}^\times$ that are squares in F. Then $F = \mathbb{Q}(\sqrt{d} : d \in D)$.*

Proposition 2.5 (Unramified quadratic extensions [Sil09, proof of Proposition VIII.1.6])**.** *Let p be an odd prime and $d \in \mathbb{Q}^\times$. Then $\mathbb{Q}(\sqrt{d})$ is unramified at p if and only if $v_p(d)$ is even.*

Definition 2.6 (The group $\mathbb{Q}(S, 2)$ [Sil09, Theorem X.1.1(c)])**.** The subgroup $\mathbb{Q}(S, 2) \subseteq \mathbb{Q}^\times/\mathbb{Q}^{\times 2}$ consists of the classes of the $d \in \mathbb{Q}^\times$ with $v_p(d)$ even for every prime $p \notin S$. It is generated by the classes of -1 and of the primes in S, so it has order 2^{s+1}.

Problem 2.7 ($L/\mathbb{Q}$ is unramified outside S)**.**

(1) Show that $\mathrm{Gal}(L/\mathbb{Q})$ is abelian of exponent dividing 2. Let $D \subseteq \mathbb{Q}^\times/\mathbb{Q}^{\times 2}$ be the subgroup of the classes of the $d \in \mathbb{Q}^\times$ that are squares in L. Deduce that $L = \mathbb{Q}(\sqrt{d} : d \in D)$.
(2) Let $p \notin S$. Show that E has good reduction at p, and that O, T_1, T_2, T_3 reduce to four distinct points of $E_p(\mathbb{F}_p)$. Deduce that reduction is injective on $E[2](\overline{\mathbb{Q}})$.
(3) Let $p \notin S$, $Q \in E(\overline{\mathbb{Q}})$ with $2Q \in E(\mathbb{Q})$, and $\sigma \in I_p$. Show that $\sigma(Q) - Q$ reduces to $\bar{O}$, and deduce that $\sigma(Q) = Q$. Conclude that $L/\mathbb{Q}$ is unramified at p.
(4) Show that $D \subseteq \mathbb{Q}(S, 2)$. Deduce that $\#\,\mathrm{Gal}(L/\mathbb{Q}) \leqslant 2^{s+1}$ and $\#E(\mathbb{Q})/2E(\mathbb{Q}) \leqslant 4^{s+1}$.

3. An example

This section runs Problem 2.7 on $y^2 = x^3 - x$, that is, $(e_1, e_2, e_3) = (0, 1, -1)$. The images of its points of order two under the explicit form of the Kummer map are [Sil09, Proposition X.1.4].

Problem 3.1 (The curve $y^2 = x^3 - x$)**.**
In this problem, E is the curve $y^2 = x^3 - x$, with $T_1 = (0, 0)$, $T_2 = (1, 0)$ and $T_3 = (-1, 0)$. `Magma` finds its rank and its torsion subgroup, and checks that the halves of each T_i are defined over $\mathbb{Q}(\zeta_8)$:

```
E := EllipticCurve([-1, 0]);
Rank(E);                        // 0 true
TorsionSubgroup(E);             // Z/2 + Z/2
K<z> := CyclotomicField(8);
```

```
EK := ChangeRing(E, K);
[#DivisionPoints(EK ! [e, 0], 2) : e in [0, 1, -1]];   // [ 4, 4, 4 ]
```

So $E(\mathbb{Q}) = E[2](\overline{\mathbb{Q}})$.
(1) Compute S and the bound of Problem 2.7(4).
(2) For $i = 1, 2, 3$, find the four $Q \in E(\overline{\mathbb{Q}})$ with $2Q = T_i$, and show that $L = \mathbb{Q}(i, \sqrt{2}) = \mathbb{Q}(\zeta_8)$.[a]
(3) Let $\sigma, \tau \in \mathrm{Gal}(L/\mathbb{Q})$ with $\sigma(i) = -i$, $\sigma(\sqrt{2}) = \sqrt{2}$, $\tau(i) = i$ and $\tau(\sqrt{2}) = -\sqrt{2}$. Compute $\kappa(T_i, \sigma)$ and $\kappa(T_i, \tau)$ for $i = 1, 2, 3$, and deduce that no T_i lies in $2E(\mathbb{Q})$. Compute $\#E(\mathbb{Q})/2E(\mathbb{Q})$ and compare it with the bound of (1).

[a] *Hint:* for $Q = (x, y)$ with $y \neq 0$, the duplication formula on this curve reads $x(2Q) = (x^2 + 1)^2/(4y^2)$.

References

[Mil22] James S. Milne, *Fields and Galois theory*, Kea Books, Ann Arbor, MI, 2022. ↑

[Sil09] Joseph H. Silverman, *The arithmetic of elliptic curves*, Second, Graduate Texts in Mathematics, vol. 106, Springer, Dordrecht, 2009. MR2514094 ↑